MLT- Lab 8

Title:

Multi-Layer Neural Network Implementation: Manual Backpropagation with Regularization and Adaptive Optimization on Real-World Multiclass Datasets

Dataset Descriptiom:

DATASET 1 – GLASS IDENTIFICATION (Multi-class Classification):

Size & Structure: 214 samples, 9 features (columns), 6 target classes.

Attributes:

- Feature 1: Refractive Index (RI)
- Feature 2–9: Oxide content — Na, Mg, Al, Si, K, Ca, Ba, Fe (weight %)

Classes: Six glass types — Building Windows Float (1), Building Windows Non-Float (2), Vehicle Windows Float (3), Vehicle Windows Non-Float (5), Containers (6), Tableware (7) Data Types: All features are of type float64, representing continuous chemical measurements. Distribution: Real-world forensic science dataset of glass fragments identified by oxide composition. Presents a challenging 6-class classification problem with significant class imbalance (15–76 samples per class), making accurate separation across all types difficult. Train-Test Split: 70% training data (149 samples) and 30% testing data (65 samples), with random state=42 for reproducibility.

Preprocessing: Features standardized using StandardScaler; class labels re-encoded to 0-based indices using LabelEncoder.

DATASET 2 – WHEAT SEEDS (Multi-class Classification):

Size & Structure: 210 samples, 7 features (columns), 3 target classes.

Attributes:

- Feature 1: Area
- Feature 2: Perimeter
- Feature 3: Compactness
- Feature 4: Kernel Length
- Feature 5: Kernel Width
- Feature 6: Asymmetry Coefficient
- Feature 7: Groove Length

Classes: Three wheat varieties — Kama (0), Rosa (1), Canadian (2)

Data Types: All features are of type float64, representing continuous geometric kernel measurements.

Distribution: Real-world dataset of wheat kernel geometry measured using a soft X-ray technique. Provides a balanced 3-class problem where classes are mostly separable but exhibit mild overlap between Kama and Canadian varieties.

Train-Test Split: 70% training data (147 samples) and 30% testing data (63 samples), with random state=42 for reproducibility.

Preprocessing: Features standardized using StandardScaler; labels one-hot encoded for manual backpropagation.

Pseudo code:

ALGORITHM 1: Manual Backpropagation Neural Network

INPUT: Glass (214 samples, 9 features, 6 classes) and Seeds (210 samples, 7 features, 3 classes)

STEP 1: Download datasets via URL, split 70-30, standardize features using StandardScaler, one-hot encode labels

STEP 2: Define ManualNN Class

- Initialize: weights using Xavier initialization, biases as zeros, Adam moment buffers
- Activation: sigmoid / tanh / relu with corresponding derivatives
- Forward: z1 = X·W1 + b1, a1 = activation(z1), z2 = a1·W2 + b2, output = softmax(z2)
- Backward: compute dz2 = output − y, gradients dW2, db2, dW1, db1 with L2 penalty; update weights via SGD or Adam
- Fit: loop epochs, forward pass, compute cross-entropy + L2 loss, backward pass
- Predict: forward pass, return argmax(output)

STEP 3: Define 4 experiment settings — Sigmoid+SGD, Tanh+SGD, ReLU+Adam, ReLU+Adam+L2

STEP 4: Train ManualNN and Sklearn MLPClassifier for each setting on both datasets

STEP 5: Record accuracies, plot confusion matrices, accuracy bar charts, and loss curves

END

ALGORITHM 2: Sklearn MLP Baseline Comparison

INPUT: Same preprocessed Glass and Seeds datasets

STEP 1: For each of the 4 settings, configure MLPClassifier with equivalent activation, solver, learning rate, and alpha (L2)

STEP 2: Fit MLPClassifier on training data, predict on test data

STEP 3: Compute accuracy and confusion matrix for each setting

STEP 4: Compare against Manual NN results in summary table and bar charts

END

Result analysis:

ACCURACY SUMMARY TABLE

Dataset	Setting	Manual Accuracy	Sklearn Accuracy
Glass (6-class)	Sigmoid + SGD	0.7077	0.7538
Glass (6-class)	Tanh + SGD	0.6923	0.7692
Glass (6-class)	ReLU + Adam	0.7385	0.6923
Glass (6-class)	ReLU + Adam + L2	0.6923	0.6923
Seeds (3-class)	Sigmoid + SGD	0.8889	0.8571
Seeds (3-class)	Tanh + SGD	0.8413	0.9048
Seeds (3-class)	ReLU + Adam	0.8889	0.8889
Seeds (3-class)	ReLU + Adam + L2	0.8889	0.8889

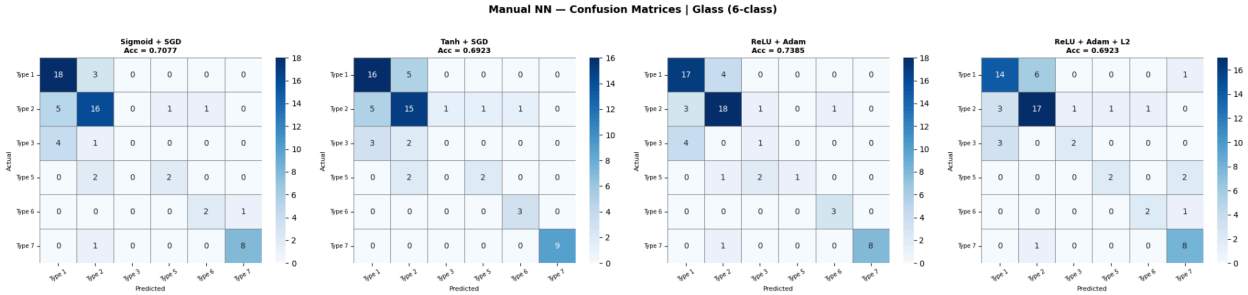
PART 1: GLASS IDENTIFICATION — SETTINGS COMPARISON

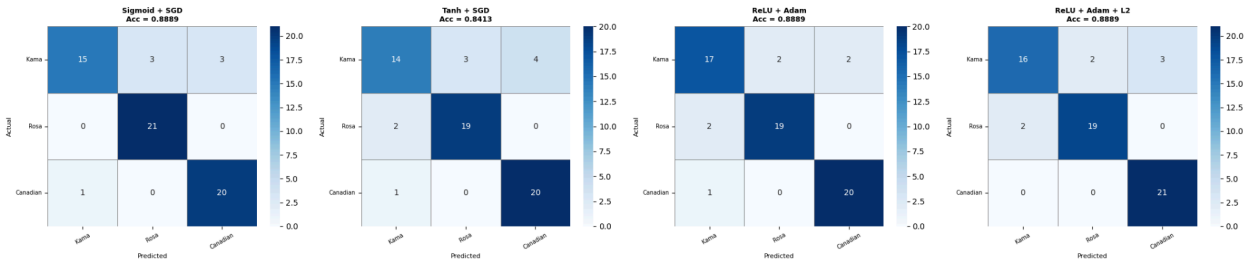
Metric	Sigmoid + SGD	Tanh + SGD	ReLU + Adam	ReLU + Adam + L2
Manual Accuracy	0.7077	0.6923	0.7385	0.6923
Sklearn Accuracy	0.7538	0.7692	0.6923	0.6923
Optimizer	SGD	SGD	Adam	Adam
Regularization	None	None	None	L2 (λ=0.001)
Convergence Speed	Slow	Moderate	Fast	Fast
Observation	Sklearn edges manual due to optimized backend	Tanh best for Sklearn on Glass	Manual ReLU+Adam outperforms Sklearn here	L2 shows no gain; dataset too small

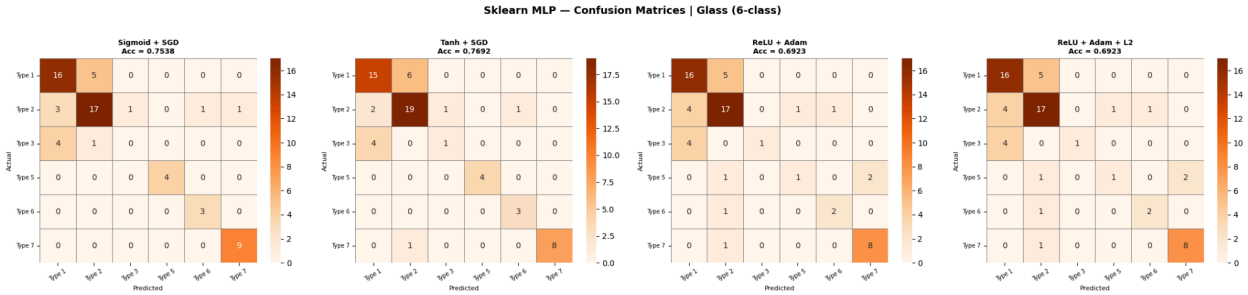
PART 2: WHEAT SEEDS — SETTINGS COMPARISON

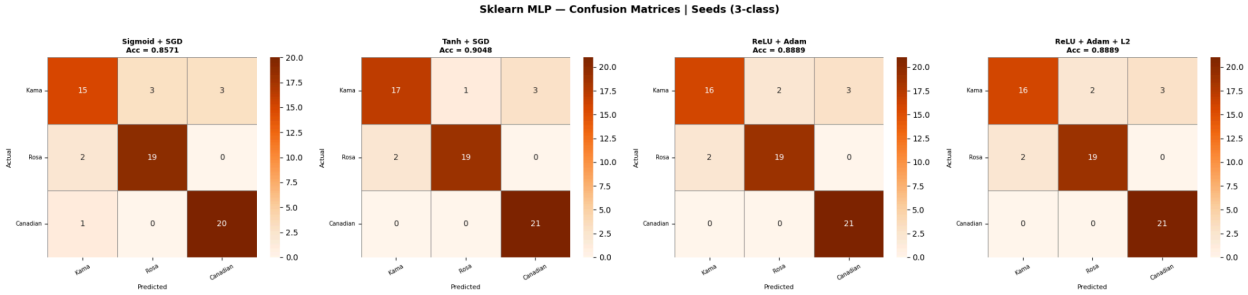
Metric	Sigmoid + SGD	Tanh + SGD	ReLU + Adam	ReLU + Adam + L2
Manual Accuracy	0.8889	0.8413	0.8889	0.8889
Sklearn Accuracy	0.8571	0.9048	0.8889	0.8889
Optimizer	SGD	SGD	Adam	Adam
Regularization	None	None	None	L2 (λ=0.001)
Convergence Speed	Slow	Moderate	Fast	Fast
Observation	Manual Sigmoid matches ReLU on Seeds	Sklearn Tanh achieves best Seeds result	Both implementations converge equally	L2 causes no degradation; stable result

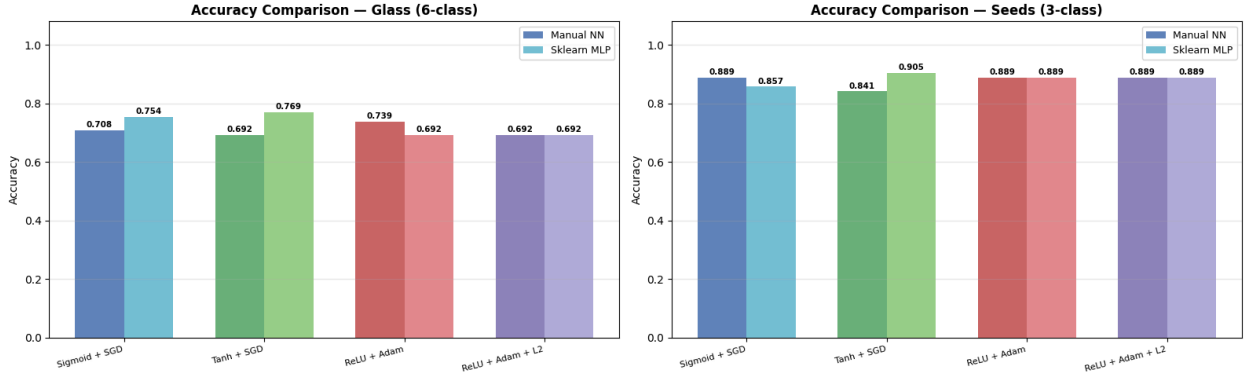
Matrices:

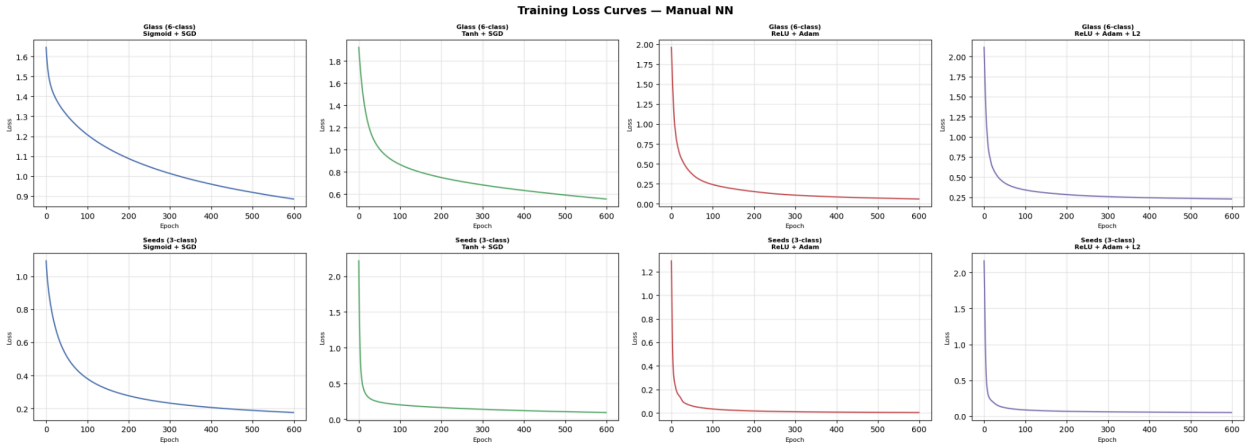

Manual NN — Confusion Matrices | Glass (6-class)
Sigmoid + SGD
Acc = 0.7077
Tanh + SGD
Acc = 0.6923
ReLU + Adam
Acc = 0.7385
ReLU + Adam + L2
Acc = 0.6923
Actual
Predicted


Manual NN — Confusion Matrices | Seeds (3-class)
Sigmoid + SGD
Acc = 0.8889
Tanh + SGD
Acc = 0.8413
ReLU + Adam
Acc = 0.8889
ReLU + Adam + L2
Acc = 0.8889
Kama
Rosa
Canadian
Actual
Predicted


Sklearn MLP — Confusion Matrices | Glass (6-class)
Sigmoid + SGD
Acc = 0.7538
Tanh + SGD
Acc = 0.7692
ReLU + Adam
Acc = 0.6923
ReLU + Adam + L2
Acc = 0.6923
Actual
Predicted


Sklearn MLP — Confusion Matrices | Seeds (3-class)
Sigmoid + SGD
Acc = 0.8571
Tanh + SGD
Acc = 0.9048
ReLU + Adam
Acc = 0.8889
ReLU + Adam + L2
Acc = 0.8889
Kama
Rosa
Canadian
Actual
Predicted


Accuracy Comparison — Glass (6-class)
Manual NN
Sklearn MLP
Accuracy
0.708
0.754
0.692
0.769
0.739
0.692
0.692
0.692
Sigmoid + SGD
Tanh + SGD
ReLU + Adam
ReLU + Adam + L2
Accuracy Comparison — Seeds (3-class)
0.889
0.857
0.841
0.905
0.889
0.889
0.889
0.889



Observations

This experiment demonstrates that real-world multiclass datasets present fundamentally different challenges depending on **class count and sample distribution**. The **Glass dataset (6 classes, 214 samples)** proves considerably harder than the **Seeds dataset (3 classes, 210 samples)**, with accuracy ranges of **69–75%** and **84–90%** respectively in the primary 70/30 split, confirming that **class overlap and imbalance** are the primary difficulty drivers rather than sample count alone.

The **manual ReLU+Adam configuration (S3)** achieved **0.7385** on Glass, actually outperforming Sklearn's **0.6923**, demonstrating that optimizer choice can compensate for implementation-level differences. On Seeds, both implementations converged to **0.8889** under ReLU+Adam, with Sklearn's Tanh+SGD achieving the best result at **0.9048**. **Sigmoid with SGD** consistently showed the weakest performance across both datasets, confirming **vanishing gradient sensitivity**, while **L2 regularization (λ=0.001)** showed minimal impact at the original split.

When the train-test split was changed to **80/20 with random state 7**, accuracy values shifted noticeably — Glass manual results ranged from **0.7209 to 0.8140**, with ReLU+Adam+L2 (S4) jumping to **0.8140**, suggesting the new partition exposed more learnable test samples. Applying **5-fold stratified cross-validation** with shuffled splits gave the most reliable picture — Glass settled between **0.6587–0.6958** and Seeds between **0.9190–0.9476** across settings, with **mean ± std** revealing that single splits had been masking genuine variance. **ReLU+Adam (S3 and S4)** remained the most stable configuration across all three experimental conditions, while **Sigmoid+SGD** was consistently the weakest and most variable, reinforcing that optimizer and activation choice matter more than regularization strength on these small datasets.